# 11.4: Cellular response: Cell signaling leads to regulation of transcription or cytoplasmic activities

We now take a closer look at the cell's subsequent response to an extracellular signal—what some researchers call the "output response." What is the nature of the final step in a signaling pathway?

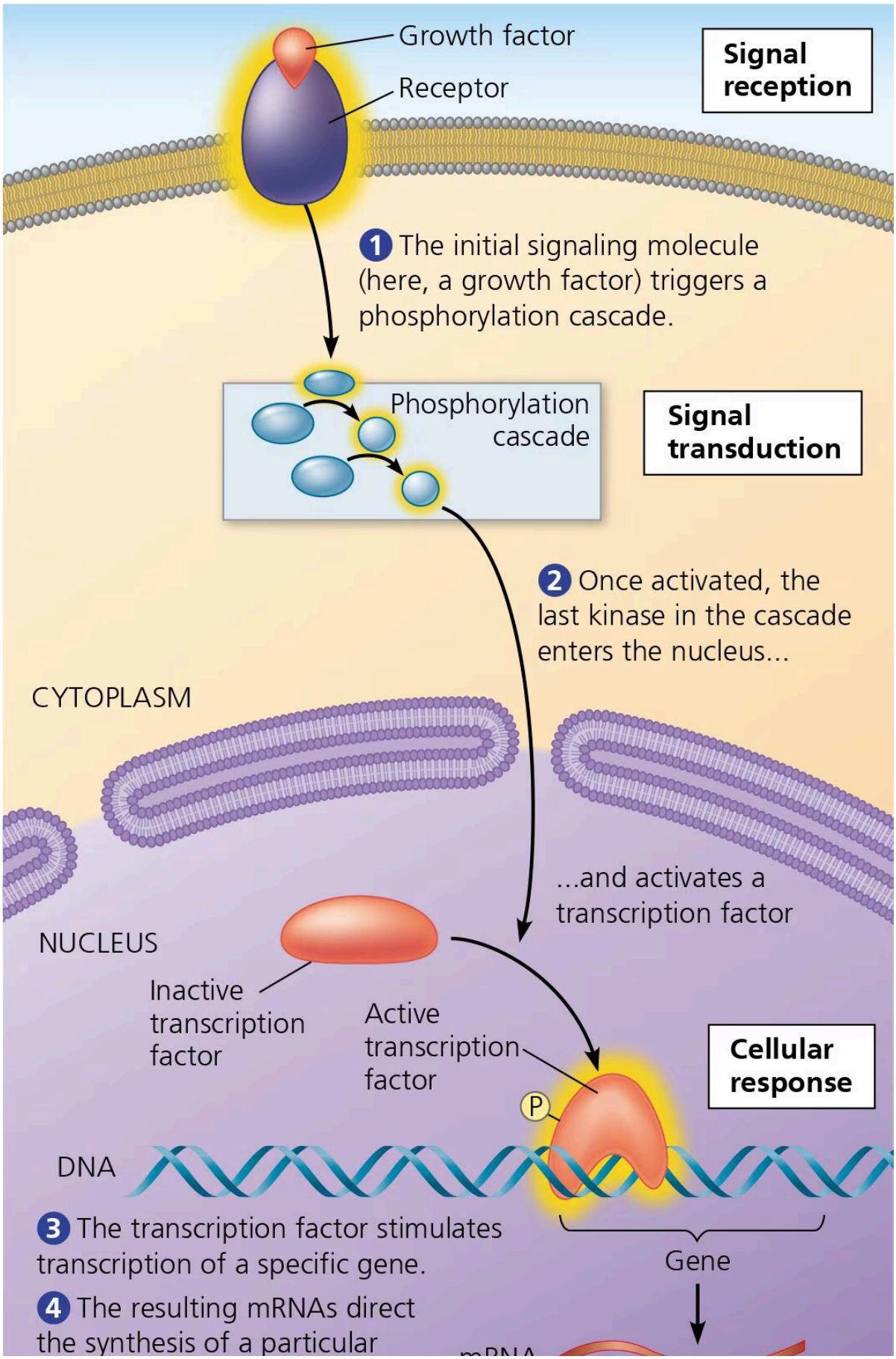
## Nuclear and Cytoplasmic Responses

Ultimately, a signal transduction pathway leads to the regulation of one or more cellular activities. The response may occur in the nucleus or in the cytoplasm of the cell.

Many signaling pathways ultimately regulate protein synthesis, usually by turning specific genes on or off in the nucleus. Like an activated steroid receptor (see Figure 11.9), the final activated molecule in a signaling pathway may function as a transcription factor. **Figure 11.15** shows an example in which a signaling pathway activates a transcription factor that turns a gene on: The response to this growth factor signal is transcription, the synthesis of one or more specific mRNAs, which will be translated in the cytoplasm into specific proteins. In other cases, the transcription factor might regulate a gene by turning it off. Often, a transcription factor regulates several different genes.

**Figure 11.15**

```
Nuclear responses to a signal: the activation of a specific gene by a growth factor.
```


Growth factor
Receptor
Signal reception
1 The initial signaling molecule (here, a growth factor) triggers a phosphorylation cascade.
Phosphorylation cascade
Signal transduction
2 Once activated, the last kinase in the cascade enters the nucleus...
CYTOPLASM
...and activates a transcription factor
NUCLEUS
Inactive transcription factor
Active transcription factor
P
Cellular response
DNA
3 The transcription factor stimulates transcription of a specific gene.
Gene
4 The resulting mRNAs direct the synthesis of a particular

Reception: a growth factor in the extracellular fluid binds to a receptor in the plasma membrane, leading to transduction.

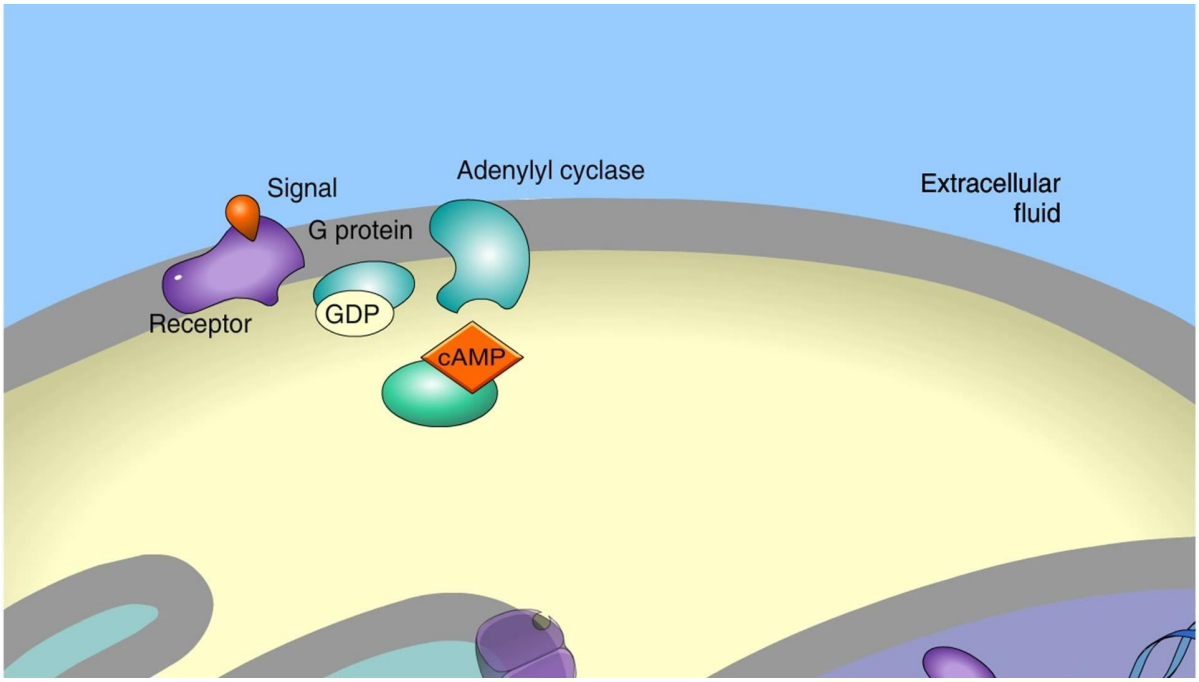
Transduction: a phosphorylation cascade occurs in the cytoplasm, leading to:

Response: An inactive transcription factor in the cell nucleus is activated (phosphorylated), causing transcription of a gene (segment of DNA) into mRNA.

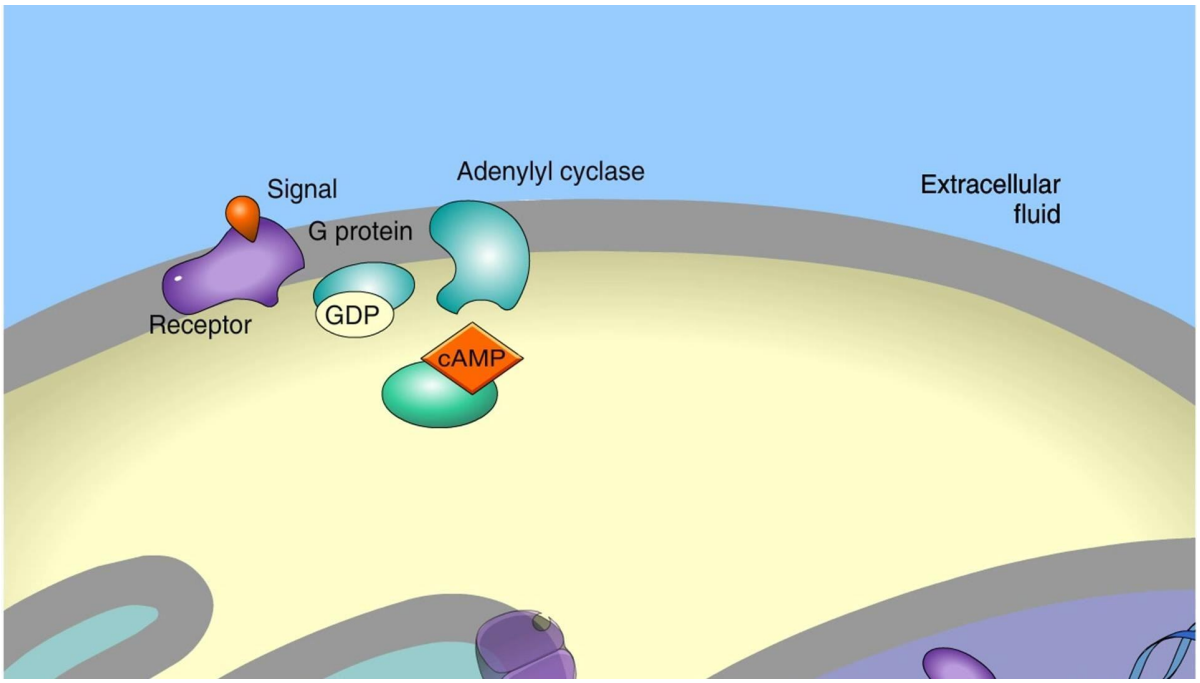
This diagram shows a typical signaling pathway that leads to regulation of gene activity in the cell nucleus. The initial signaling molecule, in this case a growth factor, triggers a phosphorylation cascade, as in Figure 11.10. (The ATP molecules and phosphate groups are not shown.) Once phosphorylated, the last kinase in the sequence enters the nucleus and activates a transcription factor, which stimulates transcription of a specific gene (or genes). The resulting mRNAs then direct the synthesis of a particular protein.

**Animation**

Nuclear Response: Activating a Gene




Play



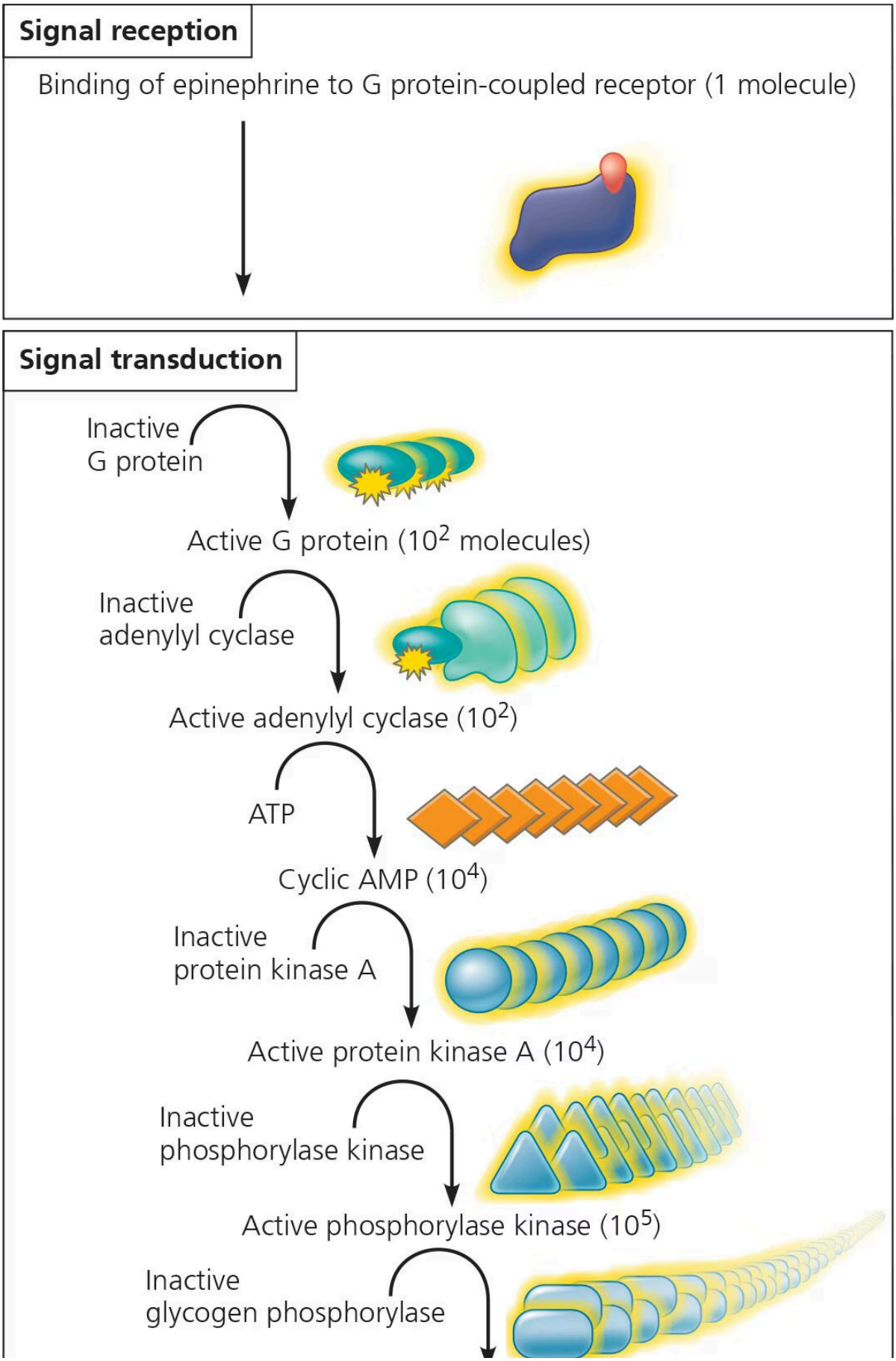

Play

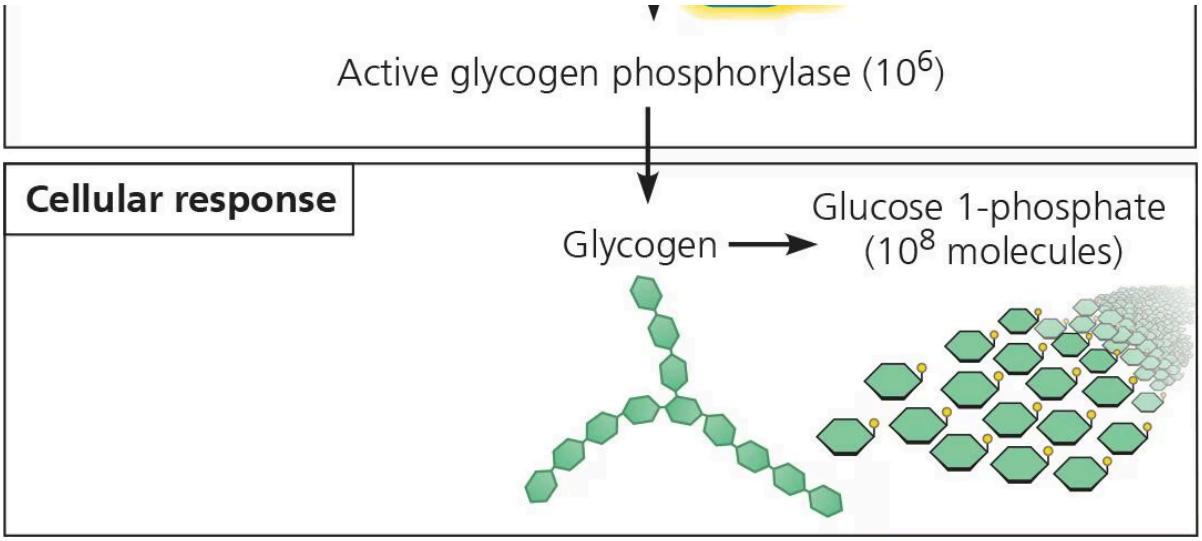
**

Sometimes a signaling pathway may regulate the *activity* of proteins rather than causing their synthesis by activating gene expression. This directly affects proteins that function outside the nucleus. For example, a signal may cause the opening or closing of an ion channel in the plasma membrane or a change in the activity of a metabolic enzyme. As we have seen, the response of liver cells to the hormone epinephrine helps regulate cellular energy metabolism by affecting the activity of an enzyme. The final step in the epinephrine signaling pathway activates the enzyme that catalyzes the breakdown of glycogen. **Figure 11.16** shows the complete pathway leading to the release of glucose 1-phosphate molecules from glycogen. Notice that as each molecule is activated, the response is amplified, a subject we'll return to shortly.

**Figure 11.16**

```
    Cytoplasmic response to a signal: the stimulation of glycogen breakdown by epinephrine
```


Signal reception
Binding of epinephrine to G protein-coupled receptor (1 molecule)
Signal transduction
Inactive G protein
Active G protein ($10^2$ molecules)
Inactive adenylyl cyclase
Active adenylyl cyclase ($10^2$)
ATP
Cyclic AMP ($10^4$)
Inactive protein kinase A
Active protein kinase A ($10^4$)
Inactive phosphorylase kinase
Active phosphorylase kinase ($10^5$)
Inactive glycogen phosphorylase




Reception: epinephrine binds to G protein coupled receptor (1 molecule).

Transduction: Inactive G protein is activated to active G protein (10 to the power of 2 molecules. Inactive adenylyl cyclase is activated to active adenylyl cyclase (10 to the power of 2 molecules). ATP goes to Cyclic AMP (10 to the power of 4 molecules). Inactive protein kinase A is activated to active protein kinase A (10 to the power of 4 molecules). Inactive phosphorylase kinase is activated to active phosphorylase kinase (10 to the power of 5 molecules). Inactive glycogen phosphorylase is activated to active glycogen phosphorylase (10 to the power of 6 molecules).

Response: Glycogen goes to Glucose 1-phosphate (10 to the power of 8 molecules).

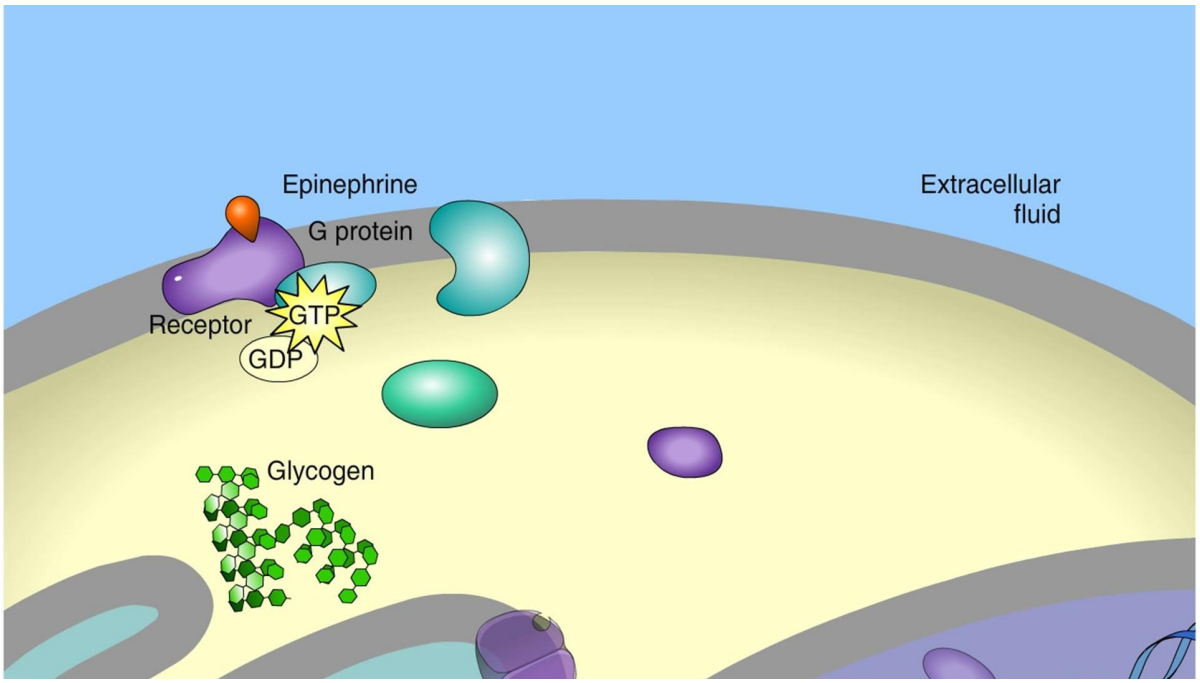
In this signaling system, the hormone epinephrine acts through a G protein-coupled receptor to activate a succession of relay molecules, including cAMP and two protein kinases (see also Figure 11.12). The final protein activated is the enzyme glycogen phosphorylase, which uses inorganic phosphate to release glucose monomers from glycogen in the form of glucose 1-phosphate molecules. This pathway amplifies the hormonal signal: One receptor protein can activate approximately 100 molecules of G protein, and each enzyme in the pathway, once activated, can act on many molecules of its substrate, the next molecule in the cascade. The number of activated molecules given for each step is approximate.

**VISUAL SKILLS** *In the figure, how many glucose*1-phosphate*molecules are released in response to one signaling molecule? Calculate the factor by which the response is amplified in going from each step to the next.*

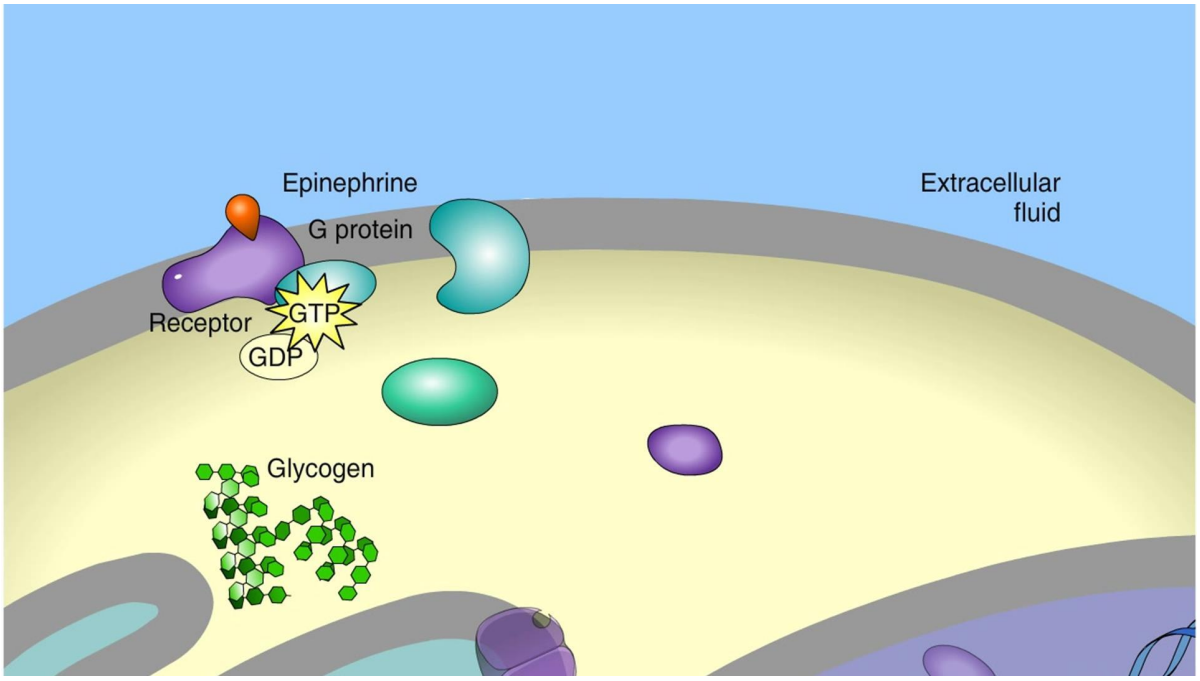
*For suggested answer, see*Appendix A.*

**Animation**

Cytoplasmic Response: Glycogen Breakdown




Play




Play

**

Signal receptors, relay molecules, and second messengers participate in a variety of pathways, leading to both nuclear and cytoplasmic responses, including cell division. Malfunctioning of growth factor pathways like the one in Figure 11.15 can contribute to abnormal cell division and the development of cancer, as we'll

see in Concept 18.5.

## Regulation of the Response

Whether the response occurs in the nucleus or in the cytoplasm, it is not simply turned “on” or “off.” Rather, the extent and specificity of the response are regulated in multiple ways. Here we’ll consider four aspects of this regulation. First, as mentioned earlier, signaling pathways generally amplify the cell’s response to a single signaling event. The degree of amplification depends on the function of the specific molecules in the pathway. Second, the many steps in a multistep pathway provide control points at which the cell’s response can be further regulated, contributing to the specificity of the response and allowing coordination with other signaling pathways. Third, the overall efficiency of the response is enhanced by the presence of proteins known as scaffolding proteins. Finally, a crucial point in regulating the response is the termination of the signal.

### Signal Amplification

Elaborate enzyme cascades amplify the cell’s response to a signal. At each catalytic step in the cascade, the number of activated products can be much greater than in the preceding step. For example, in the epinephrine-triggered pathway in Figure 11.16, each adenylyl cyclase molecule catalyzes the formation of 100 or so cAMP molecules, each molecule of protein kinase A phosphorylates about 10 molecules of the next kinase in the pathway, and so on. The amplification effect stems from the fact that these proteins persist in their active form long enough to process multiple molecules of substrate before they become inactive again. As a result of the signal’s amplification, a small number of epinephrine molecules binding to receptors on the surface of a liver cell or muscle cell can lead to the release of hundreds of millions of glucose molecules from glycogen.

### The Specificity of Cell Signaling and Coordination of the Response

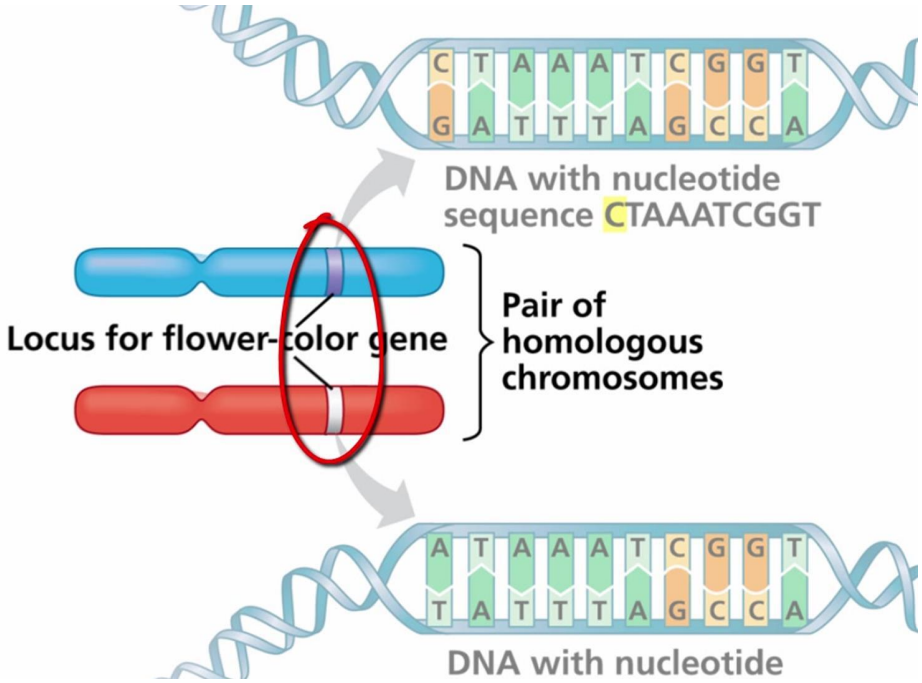
Consider two different cells in your body—a liver cell and a heart muscle cell, for example. Both are in contact with your bloodstream and are therefore constantly exposed to many different hormone molecules, as well as to local regulators secreted by nearby cells. Yet the liver cell responds to some signals but ignores others, and the same is true for the heart cell. And some kinds of signals trigger responses in both cells—but different responses. For instance, epinephrine stimulates the liver cell to break down glycogen, but the main response of the heart cell to epinephrine is contraction, leading to a more rapid heartbeat. How do we account for this difference?

The explanation for the specificity exhibited in cellular responses to signals is the same as the basic explanation for virtually all differences between cells: Because different kinds of cells turn on different sets of genes, *different kinds of cells have different collections of proteins*. The response of a cell to a signal depends on its particular collection of signal receptor proteins, relay proteins, and proteins needed to carry out the response. A liver cell, for example, is poised to respond appropriately to epinephrine by having the proteins listed in Figure 11.16 as well as those needed to manufacture glycogen.

Thus, two cells that respond differently to the same signal differ in one or more proteins that respond to the signal. Also, within some cells, there are more complex pathways that branch or converge, as shown in **Figure 11.17**. Notice, though, that these different pathways still have some molecules in common. For example, cells A, B, and C all use the same receptor protein for the red signaling molecule; differences in other proteins account for their differing responses. In cell D, a different receptor protein is used for the same signaling molecule, leading to yet another response. In cell B, a pathway triggered by one signal diverges to produce two responses; such branched pathways often involve receptor tyrosine kinases (which can activate multiple relay proteins) or second messengers (which can regulate numerous proteins). In cell C, two pathways triggered by separate signals converge to modulate a single response. Branching of pathways and “cross-talk” (interaction) between pathways are important in regulating and coordinating a cell’s responses to information coming in from different sources in the body. (You’ll learn more about this coordination in Concept 11.5.) Moreover, the use of some of the same proteins in more than one pathway allows the cell to economize on the number of different proteins it must make.

**Figure 11.17**

```
The specificity of cell signaling.
```




*The particular proteins a cell possesses determine what signaling molecules it responds to and the nature of the response. The four cells in these diagrams respond to the same signaling molecule (red) in different ways because each has a different set of proteins (purple and teal). Note, however, that the same kinds of molecules can participate in more than one pathway.*

Cell A: Pathway leads to a single response. Signaling molecule in the extracellular fluid binds to receptor in the plasma membrane, triggering a pathway of relay molecules inside the cell that lead to a single Response 1.

. Cell B: Pathway branches to two responses. The same signal molecule shown in cell A binds to the same receptor in a cell membrane, but this time the relay molecule pathway branches, leading to two different responses (responses 2 and 3).

Cell C: Cross-talk occurs between two pathways. Signal molecule from cells A and B binds again to the same membrane receptor, triggering response 4 following a relay molecule pathway inside the cell. This time, there is also another receptor in the plasma membrane that binds a different signal molecule; the result of this pathway in activation of inhibition of the pathway leading to response 4 in the cell.

Cell D: Different receptor leads to a different response. The same signal molecule shown in cells A, B and C binds to a different receptor in the plasma membrane, activating a relay molecule pathway that leads to response 5.

**VISUAL SKILLS** *Study the signaling pathway shown in*Figure 11.14*, and explain how the situation pictured for cell B in***Figure 11.17**could apply to that pathway.*

*For suggested answer, see*Appendix A.*

**Instructors:** The Scientific Skills Exercise "Using Experiments to Test a Model" can be assigned in **Mastering Biology**. It explores the cellular response of a yeast cell to the signal initiated by a mating factor from a cell of the opposite mating type.

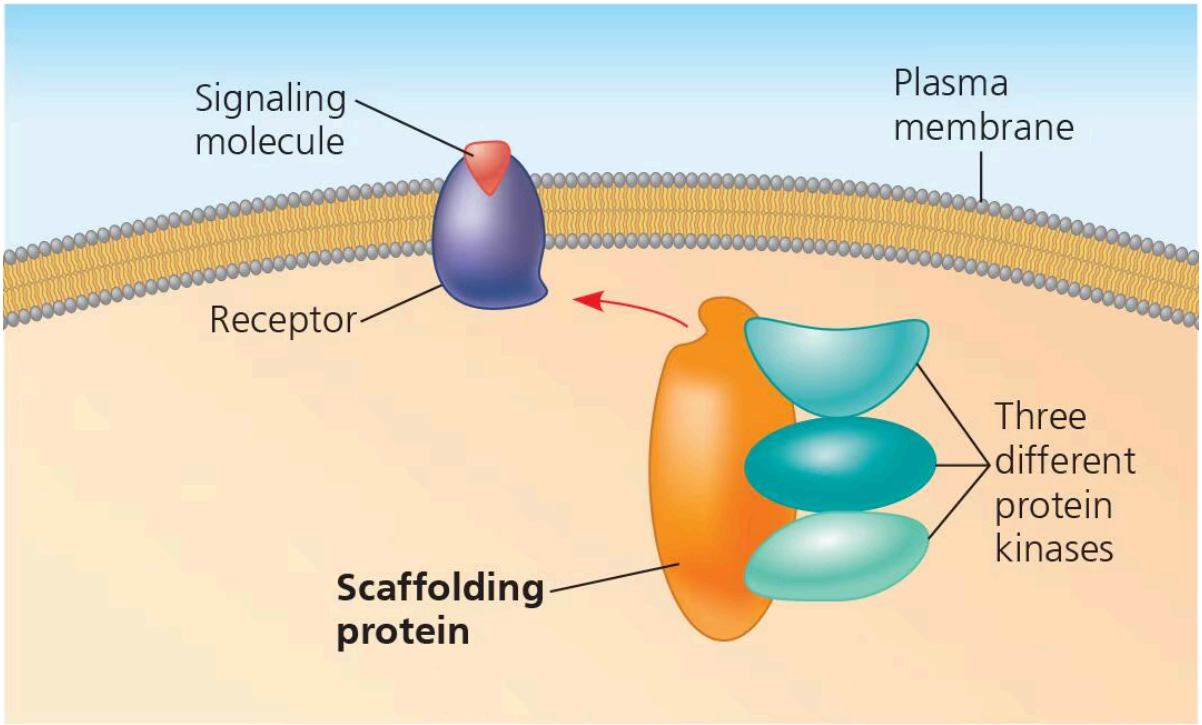
## Signaling Efficiency: Scaffolding Proteins and Signaling Complexes

The illustrations of signaling pathways in Figure 11.17 (as well as diagrams of other pathways in this chapter) are greatly simplified. The diagrams show only a few relay molecules and, for clarity's sake, display these molecules spread out in the cytosol. If this were true in the cell, signaling pathways would operate very inefficiently because most relay molecules are proteins, and proteins are too large to diffuse quickly through the viscous cytosol. How does a given protein kinase, for instance, find its protein substrate?

In many cases, the efficiency of signal transduction is apparently increased by the presence of **scaffolding proteins**, large relay proteins to which several other relay proteins are simultaneously attached **(Figure 11.18)**. Researchers have found scaffolding proteins in brain cells that *permanently* hold together networks of signaling pathway proteins at synapses. This hardwiring enhances the speed and accuracy of signal transfer between cells because the rate of protein-protein interaction is not limited by diffusion. Furthermore, in some cases the scaffolding proteins themselves may directly activate relay proteins.

**Figure 11.18**

```
A scaffolding protein.
```



*The scaffolding protein shown here (orange) simultaneously binds to a specific activated membrane receptor and three different protein kinases. This physical arrangement facilitates signal transduction by these molecules.*

The cell has a receptor in the membrane. A signaling molecule binds with the receptor outside the membrane. Relay molecules inside the membrane lead to Response 1.

The importance of the relay proteins that serve as points of branching or intersection in signaling pathways is highlighted by the problems arising when these proteins are defective or missing. For instance, in an inherited disorder called Wiskott-Aldrich syndrome (WAS), the absence of a single relay protein leads to such diverse effects as abnormal bleeding, eczema, and a predisposition to infections and leukemia. These symptoms are thought to arise primarily from the absence of the protein in cells of the immune system. By studying normal cells, scientists found that the WAS protein is located just beneath the immune cell surface. The protein interacts both with microfilaments of the cytoskeleton and with several different components of signaling pathways that relay information from the cell surface, including pathways regulating immune cell proliferation. This multifunctional relay protein is thus both a branch point and an important intersection point in a complex signal transduction network that controls immune cell behavior. When the WAS protein is absent, the cytoskeleton is not properly organized and signaling pathways are disrupted, leading to the WAS symptoms.

## Termination of the Signal

In the interest of keeping Figure 11.17 simple, we did not show the *inactivation* mechanisms that are an essential aspect of any cell-signaling pathway. For a cell of a multicellular organism to remain capable of responding to incoming signals, each molecular change in its signaling pathways must last only a short time. As we saw in the cholera example, if a signaling pathway component becomes locked into one state, whether active or inactive, consequences for the organism can be serious.

The ability of a cell to receive new signals depends on reversibility of the changes produced by prior signals. The binding of signaling molecules to receptors is reversible. As the external concentration of signaling molecules falls, fewer receptors are bound at any given moment, and the unbound receptors revert to their inactive form. The cellular response occurs only when the concentration of receptors with bound signaling molecules is above a certain threshold. When the number of active receptors falls below that threshold, the cellular response ceases. Then, by a variety of means, the relay molecules return to their inactive forms: The GTPase activity intrinsic to a G protein hydrolyzes its bound GTP; the enzyme phosphodiesterase converts cAMP to AMP; protein phosphatases inactivate phosphorylated kinases and other proteins; and so forth. As a result, the cell is soon ready to respond to a fresh signal.

**BioFlix® Animation**

Mechanism of Hormone Action: Second Messenger cAMP

Play

Play

In this section, we explored the complexity of signaling initiation and termination in a single pathway, and we saw the potential for pathways to intersect with each other. In the next section, we'll consider one especially important network of interacting pathways in the cell.

---